

Solving $Ax = 0$: Pivot Variables, Special Solutions

Lesson 6 said what the nullspace is. This lesson is the recipe for finding all of it, for any matrix, even when there are more funds than rules: pivots, free variables, rank, special solutions and the reduced form R .

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. Elimination keeps the nullspace

Elimination rewrites the rules ("return beyond 12%"), not the trades: do-nothing trades survive, other answers change.

2. Warm-up: one rule, three funds

Zero-cost trades: one pivot, two free amounts, two special solutions.

3. Pivots, free variables and rank

The 2×2 twins and the 3×3 funds: which amounts you choose, which are forced, and the rank.

4. The full recipe: four funds, three rules

A 3×4 matrix to a staircase, two special solutions, every do-nothing trade, with a try-it box.

5. Counting: $n - r$ special solutions

More funds than rules always leaves a do-nothing trade.

6. The reduced form R and the nullspace matrix

Clear above the pivots too: $R = [I \ F]$, and the special solutions are read off as $N = \begin{bmatrix} -F \\ I \end{bmatrix}$.

METHODS IN THIS LESSON

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PART 2 Prerequisites

Click any link to review it, then come back.

- The nullspace and elimination on $Ax = 0$ (Lesson 6): do-nothing trades, free columns.
- All solutions = one answer + the nullspace (Lesson 6).
- Elimination, 3×3 and back substitution (Lesson 2).

- Gauss–Jordan (Lesson 3): clearing above the pivots as well as below.

PART 3 Explanation

THE WHOLE IDEA IN ONE LINE

Eliminate until the matrix is a staircase. Columns with a pivot hold the amounts that are **forced**; columns without one hold amounts you can **choose freely**. Set one free amount to 1 and the rest to 0, solve: that's a **special solution**. All do-nothing trades are mixes of the special solutions.

THE RUNNING EXAMPLE: MORE FUNDS THAN BEFORE

Rules (rows): budget (₹ lakh), return (₹ thousand), market exposure. Funds (columns), each showing what ₹1 lakh of it does:

Fund	Column (budget, return, exposure)	Note
Nifty	(1, 12, 1)	12%, beta 1
Nifty ETF	(1, 12, 1)	a twin of Nifty from another fund house
Bonds	(1, 7, 0)	7%, beta 0
Balanced	(1, 9.5, 0.5)	half Nifty, half bonds
Midcap	(1, 15, 1.2)	15%, beta 1.2

Real fund menus look like this: many products, a few things you care about, and plenty of overlap. This lesson finds every overlap systematically.

1. Elimination keeps the nullspace

THE WHOLE IDEA IN ONE LINE

Elimination doesn't touch the trades. It only **rewrites the rules** in a simpler form. A trade that changes none of the old rules changes none of the new ones, and the other way round. So the do-nothing trades stay exactly the same.

Take Lesson 6's funds: Nifty, bonds, balanced. Elimination turned A into the staircase U :

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \longrightarrow U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Before we trust U to find the do-nothing trades, let's see what it really is.

Step 1: each row is a rule, a question you ask about a trade

Each row of A asks one question about a trade $(x, y, z) = (\text{₹ lakh into Nifty, bonds, balanced})$:

Row of A	The question	Formula
1	How much does the budget change?	$x + y + z$
2	How much does the return change?	$12x + 7y + 9.5z$
3	How much does the exposure change?	$x + 0y + 0.5z$

A do-nothing trade is one where **all three answers are 0**.

Step 2: what elimination does to a rule

The first step was "row 2 $- 12 \times$ row 1". In words, it replaces the return question with a new one:

THE NEW RULE 2: "RETURN MINUS 12 $\times$ BUDGET"

How much return does the trade make **beyond** what the same money would earn in Nifty at 12%? Per lakh: Nifty $12 - 12 = 0$, bonds $7 - 12 = -5$, balanced $9.5 - 12 = -2.5$. That's exactly row 2 of U : $(0, -5, -2.5)$.

Row 3 of U is built the same way, from (exposure) $-$ (budget) $- 0.2 \times$ (new rule 2). For these funds it comes out $(0, 0, 0)$: it asks a question whose answer is **always 0**, so it carries no information. (That's because exposure was already fixed by budget and return, $R = 7B + 5E$.)

Step 3: a do-nothing trade passes the new rules too

1 THE TRADE

$(1, 1, -2)$: buy ₹1 lakh Nifty and ₹1 lakh bonds, sell ₹2 lakh balanced.

OLD RULES

Budget $1 + 1 - 2 = 0$. Return $12 + 7 - 19 = 0$. Exposure $1 + 0 - 1 = 0$. All zero.

2 NEW RULE 2

"Return minus $12 \times$ budget" $= 0 - 12 \times 0 = 0$. If both old answers are 0, a combination of them must be 0.

CHECK IN U

Row 1: $1 + 1 - 2 = 0$. Row 2: $0(1) - 5(1) - 2.5(-2) = -5 + 5 = 0$. Row 3: 0. All zero ✓.

Step 4: and no new trade sneaks in

Could U have extra do-nothing trades that A doesn't? No, because the old rules can be rebuilt from the new ones: return $=$ (new rule 2) $+ 12 \times$ budget. If a trade gives 0 for the new rules, it gives $0 + 12 \times 0 = 0$ for the old return rule too. Nothing is lost and nothing is added.

RULE · ROW OPERATIONS KEEP THE NULLSPACE

Subtracting a multiple of one row from another, swapping two rows, or scaling a row by a non-zero number can always be undone. So $Ax = 0$ and $Ux = 0$ (and later $Rx = 0$) have exactly the same solutions.

Step 5: but the answers for other trades do change

Feed in trades that aren't do-nothing and compare what A and U report:

Trade	A reports (budget, return, exposure)	U reports (budget, return beyond 12%, always-0)
Buy ₹1L Nifty $(1, 0, 0)$	$(1, 12, 1)$	$(1, 0, 0)$
Buy ₹1L bonds $(0, 1, 0)$	$(1, 7, 0)$	$(1, -5, 0)$
₹7L Nifty + ₹3L bonds $(7, 3, 0)$	$(10, 105, 7)$	$(10, -15, 0)$
Do-nothing $(1, 1, -2)$	$(0, 0, 0)$	$(0, 0, 0)$

Different questions give different answers, so the **targets** U can reach are not the targets A can reach. Everything U produces has third entry 0. But A reaches $(10, 105, 7)$, whose third entry is 7. So $C(U) \neq C(A)$. Only the last row of the table, all zeros, is the same for both.

CAREFUL

Use U (or R) to find the **nullspace**. To test whether a target is reachable, never look at U 's columns. Carry the target along through elimination instead (Lesson 8).

IN ONE LINE

Elimination rewrites the questions, not the trades: "all answers are zero" survives the rewrite, but the other answers (and so the column space) don't.

2. Warm-up: one rule, three funds

Start with the simplest possible case: a single rule. With Nifty, bonds and midcap, a **zero-cost** trade has $x + y + z = 0$, a 1×3 system $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} x = 0$.

1 PIVOT

The 1 in column 1 is the only pivot. Nothing to eliminate: one row is already a staircase.

FREE

Columns 2 and 3 have no pivot, so the bond amount y and the midcap amount z are **free**: pick them however you like.

FORCED

The Nifty amount is then forced: $x = -y - z$.

2 SPECIAL 1

$y = 1, z = 0: x = -1 - 0 = -1$. So $s_1 = (-1, 1, 0)$: sell ₹1 lakh Nifty, buy ₹1 lakh bonds.

SPECIAL 2

$y = 0, z = 1: x = -0 - 1 = -1$. So $s_2 = (-1, 0, 1)$: sell ₹1 lakh Nifty, buy ₹1 lakh midcap.

3 EVERYTHING

Every zero-cost trade is $y s_1 + z s_2$. Example: Lesson 5's trade $(-6, 1, 5) = 1 \times (-1, 1, 0) + 5 \times (-1, 0, 1) = (-1 - 5, 1 + 0, 0 + 5) \checkmark$.

MEANING

Any rebalancing that doesn't add or withdraw money is "switch some Nifty into bonds" plus "switch some Nifty into midcap". Two building blocks, because two amounts are free.

MIT's recitation does the same with $x - 5y + 2z = 0$: pivot x , free y and z , special solutions $(5, 1, 0)$ and $(-2, 0, 1)$. Check: $5 - 5(1) + 2(0) = 0 \checkmark$ and $-2 - 5(0) + 2(1) = 0 \checkmark$.

IN ONE LINE

One rule on three funds: one forced amount, two free amounts, two special solutions, a plane of do-nothing trades.

3. Pivots, free variables and rank

RULE · PIVOT VARIABLES AND FREE VARIABLES

After elimination, a **pivot column** has a pivot (the first non-zero entry of some row); its unknown is a **pivot variable**, forced by back substitution. A **free column** has no pivot; its unknown is a **free variable**, chosen freely.

DEFINITION · RANK R = THE NUMBER OF PIVOTS

The **rank** of A is how many pivots elimination finds. In money terms: how many genuinely different directions the funds give you.

2 × 2: the twins

$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$ (budget and return of Nifty and its twin ETF). Row 2 $- 12 \times$ row 1: $12 - 12 = 0$, $12 - 12 = 0$. One pivot (column 1), one free column (column 2). **Rank 1**. Special solution: ETF = 1, then $x + 1 = 0$, so $(-1, 1)$: swap ₹1 lakh of Nifty for the ETF.

3 × 3: three funds, three rules

Funds	Staircase U	Pivots in columns	Free	Rank	Special solutions
Nifty, bonds, balanced	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$	1, 2	3 (balanced)	2	$(-0.5, -0.5, 1)$
Nifty, bonds, midcap	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}$	1, 2, 3	none	3	none: $N = \{0\}$

The balanced row, entry by entry (from Lesson 6): free $z = 1$. Row 2: $-5y - 2.5(1) = 0 \Rightarrow y = -0.5$. Row 1: $x + (-0.5) + 1 = 0 \Rightarrow x = -0.5$. This is Lesson 6's $(1, 1, -2)$ scaled by $-\frac{1}{2}$: by convention a special solution sets its free variable to exactly 1.

IN ONE LINE

Pivot = forced, free = your choice; rank = number of pivots = number of truly different funds.

4. The full recipe: four funds, three rules

Now a menu with overlap in two places: Nifty, Nifty ETF, bonds, balanced. Four unknowns $(x_1, x_2, x_3, x_4) = ₹ \text{ lakh in each}$, three rules:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array}$$

This has exactly the shape of Strang's lecture example $\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$: a 3×4 matrix whose free columns turn out to be 2 and 4.

Step 1: eliminate to a staircase

METHOD · ECHELON FORM U (THE STAIRCASE)

Eliminate column by column. If a column has no non-zero entry at or below the current row, **skip it** (it's free) and move right; the pivot steps down only when one is found. The result: each row starts further right than the one above, and zero rows sit at the bottom.

1 WHAT WE DO

ROW 2

Clear column 1 below the pivot 1.

Multiplier $12/1 = 12$. Row 2 $- 12 \times$ row 1: $12 - 12(1) = 0$, $12 - 12(1) = 0$, $7 - 12(1) = -5$, $9.5 - 12(1) = -2.5$. Now $(0, 0, -5, -2.5)$.

ROW 3

Multiplier $1/1 = 1$. Row 3 $-$ row 1: $1 - 1 = 0$, $1 - 1 = 0$, $0 - 1 = -1$, $0.5 - 1 = -0.5$. Now $(0, 0, -1, -0.5)$.

2 COLUMN 2

WHY

Look for the next pivot in row 2, column 2: it's 0, and so is row 3's entry. **No pivot**, so column 2 (the ETF) is free. Move right.

The ETF column equals the Nifty column, so whatever cleared Nifty cleared the ETF too.

3 COLUMN 3

Row 2, column 3 is -5 : that's pivot 2. Row 3: multiplier $(-1)/(-5) = 0.2$. Row 3 $- 0.2 \times$ row 2: $0, 0, -1 - 0.2(-5) = 0$,

COLUMN 4

$$-0.5 - 0.2(-2.5) = 0. \text{ Row 3 is all zeros.}$$

No rows left below row 2 with a non-zero entry: column 4 (balanced) is free.

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{pivot columns: 1, 3 (Nifty, bonds)} \\ \text{free columns: 2, 4 (ETF, balanced)} \\ \text{rank } r = 2 \end{array}$$

Row 3 vanished because exposure was already a mix of the other two rules for these funds: it carried no new information.

Step 2: one special solution per free column

METHOD · SPECIAL SOLUTIONS

For each free variable in turn: set it to 1, set the other free variables to 0, and back-substitute for the pivot variables. Each free column gives one special solution.

1 FREE CHOICE

$$\text{ETF } x_2 = 1, \text{ balanced } x_4 = 0.$$

ROW 2

$$-5x_3 - 2.5(0) = 0 \Rightarrow x_3 = 0.$$

ROW 1

$$x_1 + 1 + 0 + 0 = 0 \Rightarrow x_1 = -1.$$

RESULT

$$s_1 = (-1, 1, 0, 0): \text{ sell ₹1 lakh Nifty, buy ₹1 lakh ETF.}$$

2 FREE CHOICE

$$\text{ETF } x_2 = 0, \text{ balanced } x_4 = 1.$$

ROW 2

$$-5x_3 - 2.5(1) = 0 \Rightarrow -5x_3 = 2.5 \Rightarrow x_3 = -0.5.$$

ROW 1

$$x_1 + 0 + (-0.5) + 1 = 0 \Rightarrow x_1 = -0.5.$$

RESULT

$$s_2 = (-0.5, 0, -0.5, 1): \text{ buy ₹1 lakh balanced, sell ₹0.5 lakh each of Nifty and bonds.}$$

3 CHECK S_1

$$\text{Budget } -1 + 1 + 0 + 0 = 0 \checkmark. \text{ Return } -12 + 12 + 0 + 0 = 0 \checkmark.$$

$$\text{Exposure } -1 + 1 + 0 + 0 = 0 \checkmark.$$

CHECK S_2

$$\begin{aligned} \text{Budget } -0.5 + 0 - 0.5 + 1 &= 0 \checkmark. \text{ Return } 12(-0.5) + 12(0) + 7(-0.5) + 9.5(1) = -6 - 3.5 + 9.5 = 0 \checkmark. \\ \text{Exposure } -0.5 + 0 + 0 + 0.5 &= 0 \checkmark. \end{aligned}$$

Step 3: every do-nothing trade

$N(A)$ is all mixes $c s_1 + d s_2$: a **plane** inside $\mathbb{R}^4$ (the space of four-fund trades). Choose the ETF amount c and the balanced amount d ; the Nifty and bond amounts are forced: $x_1 = -c - 0.5d$, $x_3 = -0.5d$.

Try it: choose the free amounts

ETF amount $x_2 = c$

Balanced amount $x_4 = d$

Free (you chose): ETF $x_2 = 2$, balanced $x_4 = 2$.

Forced: Nifty $x_1 = -c - 0.5d = -3$, bonds $x_3 = -0.5d = -1$.

Trade $= 2 s_1 + 2 s_2 = (-3, 2, -1, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(-3) + 1(2) + 1(-1) + 1(2) = 0$ ✓

Return: $12(-3) + 12(2) + 7(-1) + 9.5(2) = 0$ ✓

Exposure: $1(-3) + 1(2) + 0(-1) + 0.5(2) = 0$ ✓

Nothing changes: every rule stays at 0, whatever free amounts you pick.

IN ONE LINE

Staircase $\rightarrow$ pivot and free columns $\rightarrow$ one special solution per free column $\rightarrow$ all their mixes are the nullspace.

5. Counting: $n - r$ special solutions

RULE · $N - R$ FREE VARIABLES, $N - R$ SPECIAL SOLUTIONS

With n columns and rank r : r pivot variables and $n - r$ free variables, so $n - r$ special solutions. If $n > m$ (more unknowns than equations), then $r \leq m < n$, so there is **at least one** free variable and a non-zero do-nothing trade **always** exists.

Matrix	m rules	n funds	rank r	$n - r$	Nullspace
Zero-cost $[1 \ 1 \ 1]$	1	3	1	2	a plane
Twins (2×2)	2	2	1	1	a line
Nifty, bonds, balanced	3	3	2	1	a line
Nifty, bonds, midcap	3	3	3	0	only 0
Nifty, ETF, bonds, balanced	3	4	2	2	a plane in $\mathbb{R}^4$
Nifty, bonds, midcap, balanced	3	4	3	1	a line in $\mathbb{R}^4$

The last row, worked. Nifty, bonds and midcap already give 3 pivots (topic 3), so balanced must be free. Set it to 1: the forced amounts come out as $(-0.5, -0.5, 0, 1)$, the same "build balanced yourself" trade. Check: budget $-0.5 - 0.5 + 0 + 1 = 0 \checkmark$; return $-6 - 3.5 + 0 + 9.5 = 0 \checkmark$; exposure $-0.5 + 0 + 0 + 0.5 = 0 \checkmark$.

WHY A DESK CARES

Three things you control (budget, return, exposure) and four or more products: there is **guaranteed** to be a way to switch between products that changes nothing you measure. That's exactly where a desk looks for the cheaper route, and where hidden risks (things you *don't* measure) can creep in unnoticed.

IN ONE LINE

Free variables = funds – pivots. More funds than rules $\Rightarrow$ at least one do-nothing trade.

6. The reduced form R and the nullspace matrix

THE WHOLE IDEA IN ONE LINE

Keep going after U : make each pivot 1 and clear **above** it too. Then the special solutions can be read straight off the matrix, no back substitution needed.

METHOD · REDUCED ROW ECHELON FORM R (GAUSS–JORDAN ON U)

From U : divide each pivot row by its pivot, then subtract multiples of it from the rows **above** so every pivot column is all zeros except its 1. The result $R = \text{rref}(A)$ has the same nullspace as A and U .

1 PIVOT 2 → 1

Row 2 $\div (-5)$: $0, 0, -5/(-5) = 1, -2.5/(-5) = 0.5$. Now row 2 = $(0, 0, 1, 0.5)$.

CLEAR ABOVE

Row 1 has a 1 in column 3. Row 1 $- 1 \times$ row 2: $1 - 0 = 1, 1 - 0 = 1, 1 - 1 = 0, 1 - 0.5 = 0.5$. Now row 1 = $(1, 1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + 0.5x_4 = 0 \\ \text{row 2: } x_3 + 0.5x_4 = 0 \end{array}$$

Each row now says "pivot amount = $-(\text{free amounts} \times \text{these numbers})$ ": $x_1 = -x_2 - 0.5x_4$ and $x_3 = -0.5x_4$. That's the forced-amount rule from topic 4, with no work left.

Pivot columns first: $R = [I \ F]$

List the pivot columns (1, 3) first and the free columns (2, 4) after. The pivot part becomes an identity I , and the free part is a block F :

$$\begin{array}{c|cc|cc} & x_1 & x_3 & x_2 & x_4 \\ \hline \text{row 1} & 1 & 0 & 1 & 0.5 \\ \text{row 2} & 0 & 1 & 0 & 0.5 \end{array} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$$

METHOD · NULLSPACE MATRIX $N = [-F; I]$

Stack $-F$ (the pivot amounts) on top of an identity (the free amounts, each set to 1 in turn). Its columns are exactly the special solutions, and $RN = 0$. Rows of N are in the order pivot variables, then free variables.

$$N = \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{array}{c} x_1 \\ x_3 \\ x_2 \\ x_4 \end{array} \begin{bmatrix} -1 & -0.5 \\ 0 & -0.5 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{\text{back to order } x_1, x_2, x_3, x_4} s_1 = (-1, 1, 0, 0), \quad s_2 = (-0.5, 0, -$$

1 CHECK RN

Row 1 $\cdot s_1$: $1(-1) + 1(1) + 0(0) + 0.5(0) = 0 \checkmark$. Row 1 $\cdot s_2$: $1(-0.5) + 1(0) + 0(-0.5) + 0.5(1) = 0 \checkmark$.

Row 2 $\cdot s_1$: $0 + 0 + 1(0) + 0.5(0) = 0 \checkmark$. Row 2 $\cdot s_2$: $0 + 0 + 1(-0.5) + 0.5(1) = 0 \checkmark$. Row 3 is zero, so it gives 0 for both.

WHY IT WORKS

R times a trade = $I \times (\text{pivot part}) + F \times (\text{free part})$. With pivot part $-F$ and free part I : $I(-F) + FI = -F + F = 0$.

▼ For the curious: MIT's own example, and its transpose

Strang's $A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$ reduces to $R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$: pivots in columns 1 and 3, $F = \begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$, special solutions $(-2, 1, 0, 0)$ and $(2, 0, -2, 1)$. Its transpose (a 4×3) also has rank 2, with one special solution $(-1, -1, 1)$: column 3 = column 1 + column 2. Rank is the same for A and A^T , a fact Lesson 10 explains.

IN ONE LINE

$R = [I \ F]$ (after moving pivot columns first), and the special solutions are the columns of $N = \begin{bmatrix} -F \\ I \end{bmatrix}$.

▼ Check yourself (MIT problem 7.1): find R , the rank and the special solutions of $\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$.

Row 3 $- 2 \times$ row 1: $(0, -12, -3, -21)$. Row 2 $\div 4$: $(0, 1, \frac{1}{4}, \frac{7}{4})$. Row 3 $+ 12 \times$ row 2: $(0, 0, 0, 0)$. Row 1 $- 5 \times$ row 2: $(1, 0, 7 - \frac{5}{4}, 9 - \frac{35}{4}) = (1, 0, \frac{23}{4}, \frac{1}{4})$. So $R = \begin{bmatrix} 1 & 0 & 23/4 & 1/4 \\ 0 & 1 & 1/4 & 7/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, rank 2, free columns 3 and 4, and N 's columns are $(-\frac{23}{4}, -\frac{1}{4}, 1, 0)$ and $(-\frac{1}{4}, -\frac{7}{4}, 0, 1)$. Try it in the lab on the Examples tab.